Justice-Chain SOS device: budget buying guide

For SP Road shops · 14 October 2026 · minimum parts only

Shops may not have the exact parts from the plan, so for each part this guide lists **everything that will work**, what to **refuse**, and how to check it at the counter. Only the parts the device truly needs are listed. Anything the team already owns or can borrow from the college lab should not be bought.

Prices are rough estimates in ₹; bargain and compare two or three shops.

1. What can be used (take this page to the shop)

Part	Will work	Do not buy	Check at the counter	Approx. ₹
Board	**OK:** Raspberry Pi **3B+**, **3B**, **3A+**, **4** (any RAM: 1/2/4/8 GB), **Zero 2 W** or **Zero 2 WH**	**NO:** Pi 5, Pi 400/500 (no camera port), Pi Zero / Zero W (version 1: too slow), Pi 2 or older, **Pi Pico / Pico W** (a microcontroller, not a computer), Orange Pi / Banana Pi (camera won't work with our code)	Model name is printed on the board. 40 pins present (Zero 2 W: holes only, see section 2). Sealed box or ask to see it boot	Zero 2 W 1,800–2,600 3A+ 2,800–3,800 3B/3B+ 3,000–4,800 Pi 4 3,800–5,800
Camera	**OK:** Any **Raspberry Pi camera with a ribbon cable**: 5 MP **OV5647** (cheapest, most common), 8 MP IMX219 (Camera v2 or clone), Camera Module 3. Fallback: any **USB webcam**; one **with a built-in mic** also replaces the mic below	**NO: ESP32-CAM, OV2640, OV7670**, "Arducam Mini" or any camera with loose pins: these are for Arduino/ESP32 and do **not** fit the Pi camera port	Flat ribbon cable, 15 pins wide. Listing says "for Raspberry Pi". Zero 2 W needs the **narrow Zero cable** (22-pin at one end)	OV5647 350–800 IMX219 900–2,000 Webcam 600–1,800
Microphone	**OK: Mini USB microphone** (dongle). Or a USB sound card + 3.5 mm clip mic. Or skip it if the webcam has a mic	**NO:** Sound sensor modules (**KY-037, KY-038, MAX9814, MAX4466**): the Pi has no analog input. INMP441 I2S mic (works only with extra setup)	Plugs into a normal USB port. Zero 2 W also needs a micro-USB OTG adapter	150–400 (OTG adapter 30–80)
microSD card	**OK: 16 or 32 GB**, Class 10 / A1, a known brand (SanDisk, Samsung, HP, Strontium). 16 GB is enough	**NO:** Unbranded cards, cards under 8 GB. Fake "SanDisk" cards are common: buy from a proper shop	Test it at home with the free tool **H2testw** before installing anything	16 GB 250–400 32 GB 350–550
SOS button	**OK:** Any push button: 6×6 or 12×12 mm tactile, or a panel-mount push button	**NO:** Toggle or latching switches (we need press-and-release)	Button springs back after pressing	5–30
Status LED + resistor	**OK:** Any 3 mm or 5 mm LED, any resistor from **220 Ω to 1 kΩ**	**NO:** LED without a resistor (it can burn out the LED or the pin)	—	5–15
Wires	**OK:** Mini breadboard + a few **female-to-male** jumper wires. Borrow from the lab if possible	**NO:** Buying a full kit just for wires	Female end fits on the Pi pin	60–120 (or 0 if borrowed)

2. Minimum list for each board

Buy only the list for the board you find. Power is **not** on these lists: use a phone charger and a power bank the team already has (see section 3).

Option A: Raspberry Pi 3B+, 3B or 3A+

#	Item	Qty	Approx. ₹
1	Raspberry Pi 3B+ / 3B / 3A+	1	2,800–4,800
2	Raspberry Pi camera, OV5647 5 MP with ribbon cable	1	350–800
3	Mini USB microphone	1	150–400
4	microSD card, 16 GB Class 10 (32 GB if the price is close)	1	250–550
5	Push button	2	10–60
6	LED	2	5–10
7	Resistor, 220 Ω – 1 kΩ	2	2–5
8	Mini breadboard + 6 female-to-male jumper wires (skip if borrowed)	1	60–120
	Total		**about 3,600–6,700**

- Power: a 5V 2A–2.5A micro-USB phone charger for the desk, and a power bank with 2.1A or more output.
- No heatsink or case needed for the demo.

Option B: Raspberry Pi 4 (any RAM)

#	Item	Qty	Approx. ₹
1	Raspberry Pi 4 (1 GB or 2 GB is enough)	1	3,800–5,800
2	Raspberry Pi camera, OV5647 5 MP with ribbon cable	1	350–800
3	Mini USB microphone	1	150–400
4	microSD card, 16 GB Class 10 (32 GB if the price is close)	1	250–550
5	Push button	2	10–60
6	LED	2	5–10
7	Resistor, 220 Ω – 1 kΩ	2	2–5
8	Mini breadboard + 6 female-to-male jumper wires (skip if borrowed)	1	60–120
9	Stick-on heatsink set (the Pi 4 gets hot while recording)	1	50–150
	Total		**about 4,700–7,900**

- Power: a USB-C phone charger that can give 5V 3A (most 18W+ fast chargers), and a power bank with a USB-C output.
- If the Pi shows a lightning-bolt / "under-voltage" warning, the charger is too weak.

Option C: Raspberry Pi Zero 2 W (cheapest, original plan)

#	Item	Qty	Approx. ₹
1	Raspberry Pi Zero 2 W (or Zero 2 WH)	1	1,800–2,600
2	40-pin header, soldered on by the shop or the college lab (skip for Zero 2 WH)	1	30–100
3	Pi Zero camera cable (22-pin to 15-pin), if the camera doesn't include one	1	100–200
4	Micro-USB OTG adapter (for the USB mic)	1	30–80
5	Raspberry Pi camera, OV5647 5 MP with ribbon cable	1	350–800
6	Mini USB microphone	1	150–400
7	microSD card, 16 GB Class 10 (32 GB if the price is close)	1	250–550
8	Push button	2	10–60
9	LED	2	5–10
10	Resistor, 220 Ω – 1 kΩ	2	2–5
11	Mini breadboard + 6 female-to-male jumper wires (skip if borrowed)	1	60–120
	Total		**about 2,800–4,900**

- Power: any 5V 2A micro-USB phone charger and power bank.
- Ask the shop to solder the header on the spot; it takes a few minutes.

3. Don't buy these: reuse or borrow

Item	Use instead
Official power supply	A good phone charger (5V 2A+ micro-USB for Pi 3 / Zero 2; USB-C 5V 3A for Pi 4)
Power bank	One a team member already owns (10,000 mAh is plenty)
microSD card reader	Laptop card slot, a borrowed reader, or a phone OTG reader
Case	Any small plastic box for the demo. Buy a case only at the end if money is left
Spare SD card, spare camera, fan, longer camera cable	Not needed
Keyboard, mouse, monitor, HDMI cable	Not needed: the Pi is set up over Wi-Fi from a laptop (SSH)
Breadboard and jumper wires	Borrow from the college electronics lab

4. Recommended order at the shop

- **Board:** any board from section 1 works. Compare the **totals in section 2** and pick the cheapest one the shop has: usually the Zero 2 W (if they solder the header), then a Pi 3, then a Pi 4.
- **Camera:** ask for the "Raspberry Pi 5 MP camera, OV5647". It is the cheapest and most common.
- **If no Pi camera is available:** buy a USB webcam with a built-in mic and skip the separate mic. Tell Claude, because the recording code changes slightly.
- **Buy the board and camera from the same shop** if you can, and ask if they will exchange a part that doesn't work.

5. Quick checks on the first day

- Test the microSD card with H2testw (on Windows) to catch fake cards.
- Flash Raspberry Pi OS Lite (64-bit) with Raspberry Pi Imager, setting Wi-Fi and SSH in its settings.
- Run the plan's Phase 1 test scripts: LED blinks, button press is seen, 10-second video has picture and sound.
- If anything fails, take the part back to the shop within their exchange period.